

Personal Identification Method using Active Acoustic Sensing applied to the Nose pad of Eyeglasses

Kaito Isobe

kaito.isobe@iis.ise.ritsumei.ac.jp
Ritsumeikan University
Shiga, Japan

Kazuya Murao

murao@cs.ritsumei.ac.jp
Ritsumeikan University
Shiga, Japan

ABSTRACT

Smart glasses and VR goggles are becoming more and more popular. In these products, the home screen is displayed without user authentication after the power is turned on. While using the device, the user may be asked to enter passwords when logging in to applications or making payments, and it is common to use the attached touchpad or to use a handheld controller to enter information from the software keyboard on the screen. There are also eyeglass-type devices without displays, however, they generally do not authenticate the user. In this paper, we propose a method to automatically and periodically identify individuals by applying active acoustic sensing technology with a speaker and a microphone mounted on the nose pad. In the proposed method, an acoustic signal is played from a speaker in the nose pads of an eyeglass device, and the acoustic signal that passes through the nose in contact with the device is acquired from a microphone, and the individual is recognized by analyzing the frequency characteristics of the acoustic signal. The proposed method takes advantage of the fact that the nose pad of the eyeglass device always makes natural contact with the nose of the user to be recognized, and that the shape of the user's nose has its own acoustic characteristics. We implemented a prototype eyeglass-type device equipped with a microphone and speaker, conducted experiments to evaluate the effectiveness of the proposed method that involved 11 participants, and confirmed that the proposed method can recognize individuals with an average equal error rate of 0.09. Experiments were also conducted to evaluate the effectiveness of the proposal through changes in the external and internal states of the nose.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools.**

KEYWORDS

Person identification, active acoustic sensing, glasses, nose

ACM Reference Format:

Kaito Isobe and Kazuya Murao. 2022. Personal Identification Method using Active Acoustic Sensing applied to the Nose pad of Eyeglasses. In *Proceedings*

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

UbiComp/ISWC '22 Adjunct, September 11–15, 2022, Cambridge, United Kingdom

© 2022 Association for Computing Machinery.

ACM ISBN 978-1-4503-9423-9/22/09...\$15.00

<https://doi.org/10.1145/3544793.3560400>

of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC '22 Adjunct), September 11–15, 2022, Cambridge, United Kingdom. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3544793.3560400>

1 INTRODUCTION

Many smartphones, tablets, and notebooks are equipped with biometric authentication such as face recognition and fingerprint recognition. Biometric authentication is used in various situations such as device login, application login, and payment, and can be performed more quickly than password authentication. In addition, smart glasses and head-mounted displays such as Google Glass and Oculus Quest 2 (hereinafter referred to as “glass-type devices”) are becoming popular. However, most of the glass-type devices are not equipped with rich input interface due to the shape of the glasses, and input is generally done with a simple touch panel or a software keyboard using a handheld device. Many glass-type devices display the home screen without user authentication after being turned on, and may require password input when logging in to applications or making payments, either by linking to another device or using a software keyboard displayed in the screen. Although glass-type devices are rarely shared, if a malicious third party obtains the device, they may be able to view personal and privacy information. For glass-type devices, a means of authentication at the start of use and periodically during use without burdening the user is required.

In this paper, we propose a method to identify the wearer of an glass-type device by applying active acoustic sensing technology to the nose of the wearer. Active acoustic sensing technology is an object/state recognition method that uses the unique acoustic characteristics of an object. A speaker and a microphone are attached to the nose pads of a glass-type device, acoustic signals are transmitted from the speaker to the nose of the wearer, and the signals passing through the nose are obtained with the microphone. Then, the wearer is identified by analyzing the obtained signal with machine learning algorithm. Our method takes advantage of the fact that the nose pads of the glass-type device is in constant natural contact with the nose of the user to be recognized, and that the structure of the user's nose has its own acoustic characteristics.

2 RELATED WORK

This section introduces research on person identification using biometric data and research on object and human motion recognition using acoustic sensing.

2.1 Person Identification using Biometric Information

There are many methods of person identification using biometric information. Barbosa et al. [2] proposed a biometric authentication method using the surface of fingernails. By using nail growth, the authentication method satisfies the discardability of nail (the property that nail can be discarded and renewed by cutting them), and uses fingernail images taken by an ordinary camera for authentication. This method requires the fingernail to be close to the camera, which results in unnatural movements. Stefan et al. [10] developed a biometric authentication system using bone conduction in the skull. They focused on the fact that the structure of the human head - skull, cartilage, fluid, and composition - differs from user to user, resulting in different sound transmission. The system plays white noise on the side of the head from bone conduction loudspeakers and receive the signal from a microphone. Using a combination of the characteristic frequency response of the skull and the Mel frequency cepstral, they succeeded in identifying individuals with 97% accuracy. Person authentication by a camera has also been proposed [12], however, it is difficult to capture the entire face with a camera on a glass-type device due to the structure of the device. Fadi N. shibai et al. [11] proposed three different iris image data segmentation techniques and two data coding in iris recognition and obtained a high recognition accuracy of 93.33% despite testing similar irises of the same color. One of the problems with iris recognition is the cost of installing a camera.

In contrast to these works, the proposed method uses the fact that the user always makes contact with the glass-type device, and outputs a sound in the inaudible range, which does not cause any physical and psychological burden to the user.

2.2 Recognition Method using Acoustic Sensing

Many systems using passive acoustic sensing have been proposed so far, in which sounds generated by human actions are acquired for analysis. Brian et al. [1] recognized fingertip gestures by using bone conduction sound generated by fingertip movements. Chris et al. [3] created a gesture input surface by using the sound generated when a fingernail is dragged on a textured surface. Chris et al. [4] proposed a system to estimate the tapping position of the arm and fingers by analyzing the mechanical vibrations of the body. Roderick et al. [7] proposed handheld interactive devices controlled by haptic input to classify the vibration signals generated by stroking and scratching these surfaces.

Acoustic sensing offers many possibilities for user-object interaction and research using active acoustic sensing with a speaker and microphone has been proposed. Ohno et al. [8] [9] proposed a touch recognition technique using active acoustic sensing. A contact speaker and a contact microphone are attached to an existing object, and the grasping state is recognized by acquiring the difference in the way the object is touched. Adiyani et al. [6] proposed a method to infer the contact position and gesture on the body by resonating the skin surface with low-frequency ultrasonic waves and receiving them at different points on the body. Active acoustic sensing has also been used for position estimation. Kim et al. [5] proposed a method to determine the position of a cell phone by analyzing the response of an accelerometer and a microphone using

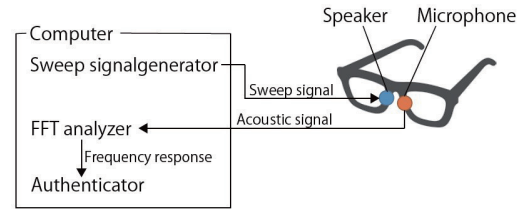


Figure 1: Overview of the proposed method.

the sound from a built-in speaker and vibration from a vibration motor.

In this paper, we propose a method of recognizing a wearer by applying active acoustic sensing to the nose pads of glasses.

3 PROPOSED METHOD

This section explains the proposed method that identifies individuals by applying active acoustic sensing technology to their noses while wearing glass-type devices.

3.1 Assumed Environment

In the proposed method, a small piezoelectric element is attached to the nose pads of a user's glass-type device as a speaker and a microphone, and acoustic signals are transmitted from the speaker. The vibration response of the nose is captured by the microphone, and the frequency response is analyzed to identify the individual. When the user wears the proposed device, nose pads keep contact with the wearer's nose while being worn. Therefore, the user is automatically authenticated when the device is worn, and the user is periodically identified while using the device. Therefore, the user does not need to perform any operation for authentication even in situations where person identification is required, such as when logging into applications or buying items.

3.2 Overview of the Proposed Method

Figure 1 shows an overview of the proposed method. Piezoelectric elements are embedded in the left and right nose pads of a glass-type device, and one is used as a speaker and the other is used as a microphone. The sweep signal generator plays a repetitive sinusoidal waveform whose frequency increases linearly within a certain period of time, called the sweep signal, from the speaker, and the response signal that passes through the nose is acquired as sound data by the microphone. The acquired sound data is subjected to FFT (Fast Fourier Transform) in the FFT analyzer to obtain the frequency power spectrum. The identification section calculates the distance between frequency power spectrum of the users registered in advance and that of the wearer to be identified.

3.3 Sweep Signal Generator

A sine wave whose frequency varies linearly from 20,001 Hz to 40,000 Hz in 0.2 second is generated as the sweep signal. This frequency range was determined by referring to previous research[8]. The audible range of human beings is approximately from 10 Hz to 20,000 Hz, depending on their age. Therefore, the sound emitted

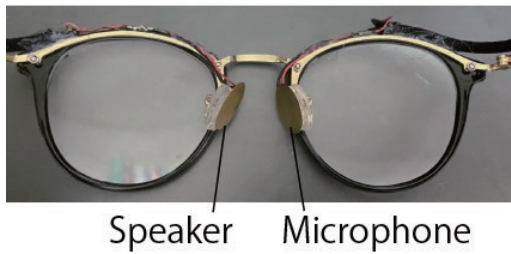


Figure 2: Glasses with a speaker and microphone.



Figure 3: A user wearing the proposed device.

from the speaker is not audible to us. The sampling frequency of the generated sweep signal is 96 kHz.

3.4 FFT Analyzer

The FFT analyzer obtains sound data from the microphone at a sampling frequency of 96 kHz and a quantization bit rate of 16 bits. The acquired sound data is cut out over a window size of 1 second in length and FFT is conducted to obtain the frequency power spectrum. 20,000 data points in the range of 20,001Hz to 40,000Hz of the power spectrum are used as a feature vector.

3.5 Authenticator

Using the obtained 20,000-dimensional feature vectors as input, person identification is performed by calculating the Euclidean distance between the user's data registered in advance and the wearer's data obtained now. If the distance is less than a threshold value, the wearer is authenticated, otherwise the wearer is rejected.

3.6 Hardware

The prototype system consists of glasses equipped with two piezo-electric elements at nose pads and a computer for signal processing and analysis. Lenovo ThinkPad X1 Carbon (Intel Core i7-10510U, RAM: 16GB, OS: Windows 10) is used as the computer for sweep signal generation, FFT analysis, and machine learning. The sound is input and output through a USB audio interface Steinberg UR22mk. The piezo electric element used is manufactured by uxcell and has a diameter of 15 mm.

The appearance of the glass is shown in Fig. 2. Piezoelectric elements are mounted on nose pads of ordinary glasses. The piezo-electric element used as a speaker is attached to the right nose pad, and the piezoelectric element used as a microphone is attached to the left nose pad. Figure 3 shows the user wearing the glasses.

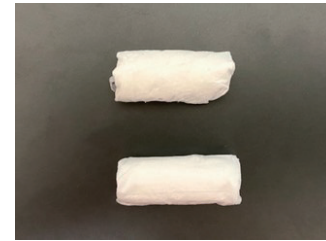


Figure 4: Cotton to stuff put into the nose.

4 EVALUATION

This section evaluates the identification accuracy of the proposed method.

4.1 Experimental Environment

Eleven male participants in their 20's (A–K) were asked to wear the prototype device, and data were collected with the following procedure. The participants wore the device while sitting on a chair, and one session of wearing the device for 10 s was conducted ten times for each participant. The participant were asked to take off the device and put it on again between sessions; therefore, the contact state between the nose pads of the device and the participant's nose changed for each session. To investigate the performance due to the change in the external condition of the nose, the same experiment was conducted by wetting the nose of five participant in their 20s (A and L–O). In addition, to evaluate the accuracy of the system due to changes in the internal condition of the nose, a similar experiment was conducted on five subjects (A and L–O) with cotton in their noses. The cotton used is shown in the Figure 4. FFT was conducted over 10 s of data per session, and the power spectra of ten samples were obtained for each participant. Random Forest was used as the machine-learning classifier, and personal-identification accuracy was evaluated through ten-segment cross-validation.

4.2 Results and Discussion

The equal error rate (EER) was used as an evaluation indicator. This rate is when the error rate of recognizing the registrant as a stranger (false rejection) and that of recognizing a stranger as a registrant (false acceptance) are equal. The EER of participants A through K when the nose was dry and that of participants A and L through O when the nose was wet, participants A and L through O with cotton in nose are listed in Tables 1, 2, and 3. When the nose was dry, the average EER was 0.09. The contact between the nose pads and nose changed as the device was removed and put on in each session. However, individuals were identified with high accuracy in a session-independent learning situation, which suggests that the acoustic properties of the human nose have the ability to identify individuals. The average EER when the nose was wet was 0.116, which is worse than the average EER of a dry nose due to the moisture on the nose surface. The average EER when cotton was placed inside the nose was 0.17, the worst accuracy among the three conditions.

In the experiments, only one type of glasses was used, and it is necessary to verify whether the proposed method works effectively

Table 1: EER of identification when participant's nose was dry.

Subject	EER
A	0.04
B	0.08
C	0.14
D	0.09
E	0.03
F	0.11
G	0.16
H	0.13
I	0.08
J	0.10
K	0.09
Average	0.09

Table 2: EER of identification when participant's nose was wet.

Subject	EER
A	0.08
L	0.13
M	0.16
N	0.09
O	0.12
Average	0.12

Table 3: EER of identification when cotton is placed in participant's nose.

Subject	EER
A	0.14
L	0.21
M	0.17
N	0.12
O	0.23
Average	0.17

with glasses made of other different materials. In addition, we conducted the experiment with the glasses-type device. It is necessary to verify whether the proposed method can identify individuals even with VR goggles. We collected data with the subjects in a natural still state. We would like to evaluate the performance while the subjects are engaged in daily activities, conversations, and work.

5 CONCLUSION

We proposed a person-identification method for eyeglass-type devices by applying active acoustic sensing technology to the user's nose. We implemented a prototype eyeglass-type device equipped with a microphone and speaker, conducted experiments to evaluate the effectiveness of the proposed method that involved 11

participant and confirmed that the proposed method can recognize individuals with an average EER of 0.09. In the future, we plan to conduct evaluations using different eyeglass-type devices, such as head-mounted displays, and under various environments.

REFERENCES

- [1] Brian Amento, Will Hill, and Loren Terveen. 2002. The sound of one hand: a wrist-mounted bio-acoustic fingertip gesture interface. In *Proceedings of the 2002 ACM SIGCHI International Conference on Human Factors in Computing Systems (CHI'02)* (2002), 724–725.
- [2] Igor Barros Barbosa, Theoharis Theoharis, and Ali E. Abdallah. 2016. On the use of fingernail images as transient biometric identifiers. *Machine Vision and Applications* 27, 1 (2016), 65–76.
- [3] Chris Harrison and Scott E. Hudson. 2008. Scratch Input, Creating Large Inexpensive, Unpowered and Mobile Finger Input Surfaces. In *Proceedings of the 21th annual ACM symposium on User interface software and technology (UIST'08)* (2008), 205–208.
- [4] Chris Harrison, Desney Tan, and Dan Morris. 2010. Skinput: appropriating the body as an input surface. In *Proceedings of the 2010 ACM SIGCHI International Conference on Human Factors in Computing Systems (CHI'10)* (2010), 453–462.
- [5] Kai Kunze and Paul Lukowicz. 2007. Symbolic Object Localization Through Active Sampling of Acceleration and Sound Signatures. In *proceedings of the 9th International Conference on Ubiquitous Computing (UbiComp'07)* (2007), 163–180.
- [6] Adiyana Mujibiya, Xiang Cao, Desney S. Tan, Dan Morris, Shwetak N. Patel, and Jun Rekimoto. 2013. The sound of touch: on-body touch and gesture sensing based on transdermal ultrasound propagation. In *Proceedings of the 2013 ACM international conference on Interactive tabletops and surfaces (ITS'13)* (2013), 189–198.
- [7] oderick Murray-Smith, John Williamson, Stephen Hughes, and Torben Quaade. 2008. Stane: synthesized surfaces for tactile input. In *Proceedings of the 2008 ACM SIGCHI International Conference on Human Factors in Computing Systems (CHI'08)* (2008), 1299–1302.
- [8] Makoto Ono, Buntarou Shizuki, and Jiro Tanaka. 2013. Touch & Activate: Adding Interactivity to Existing Objects using Active Acoustic Sensing. In *Proceedings of the 26th annual ACM symposium on User interface software and technology, UIST'13* (October 2013), 31–40.
- [9] Makoto Ono, Buntarou Shizuki, and Jiro Tanaka. 2013. Touch and Activate: Adding Interactivity to Existing Objects using Active Acoustic Sensing. In *Proceedings of the 26th annual ACM symposium on User interface software and technology (UIST'13)* (2013), 31–40.
- [10] Stefan Schneegass, Youssef Oualil, and Andreas Bulling. 2016. SkullConduct: Biometric User Identification on Eyewear Computers Using Bone Conduction Through the Skull. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI'16)* (2016).
- [11] Fadi N. Sibai, Hafsa I. Hosani, Raja M. Naqbi, and Salima Dhanhani. 2011. Iris recognition using artificial neural networks. *Expert Systems with Applications* 38, 5 (2011), 5940–5946.
- [12] Byeongtak Song and S.Ozawa. 1994. Personal identification by image-processing sequential images of human-face. *Proceedings of IECON'94 - 20th Annual Conference of IEEE Industrial Electronics* (1994).